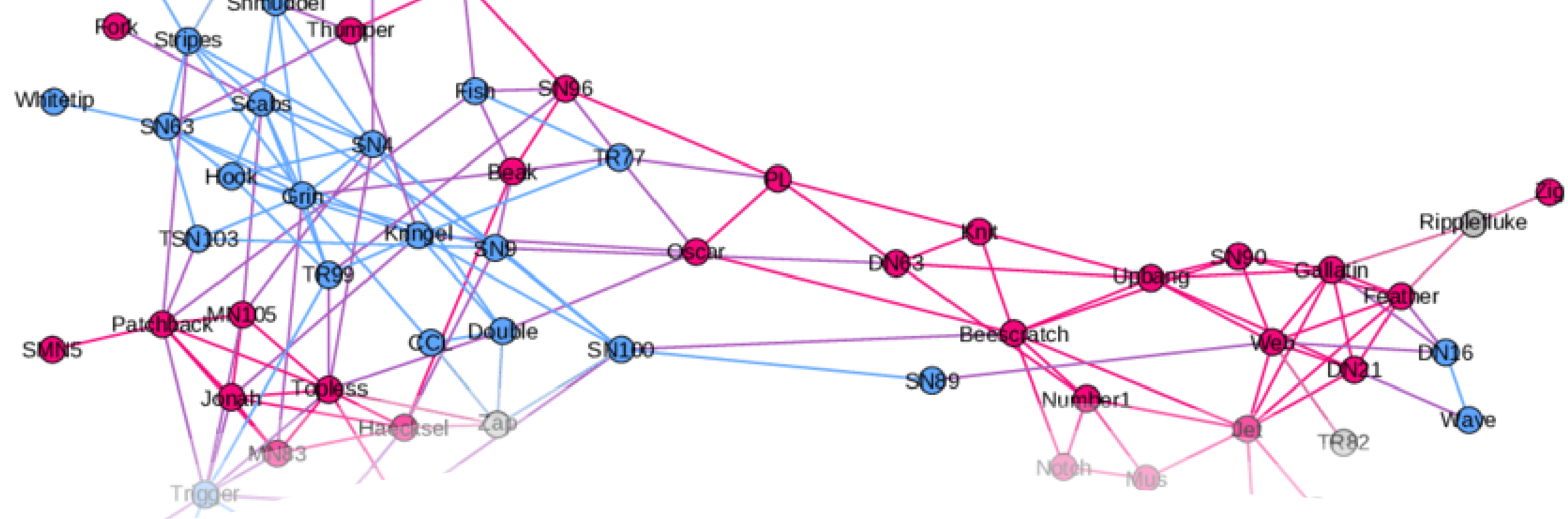
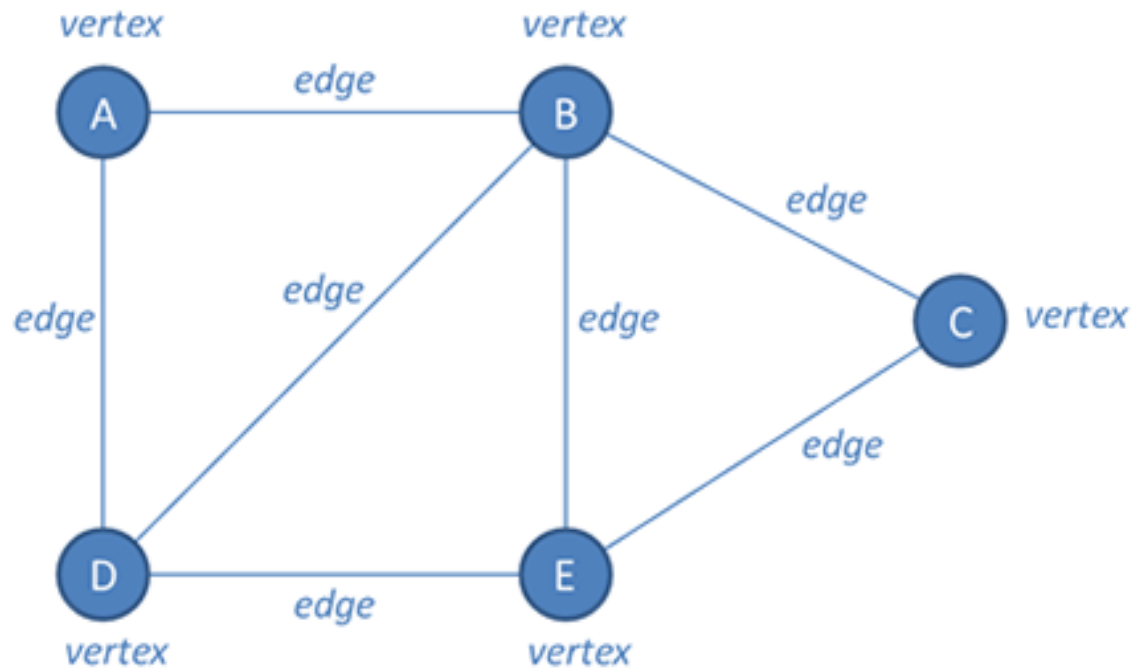


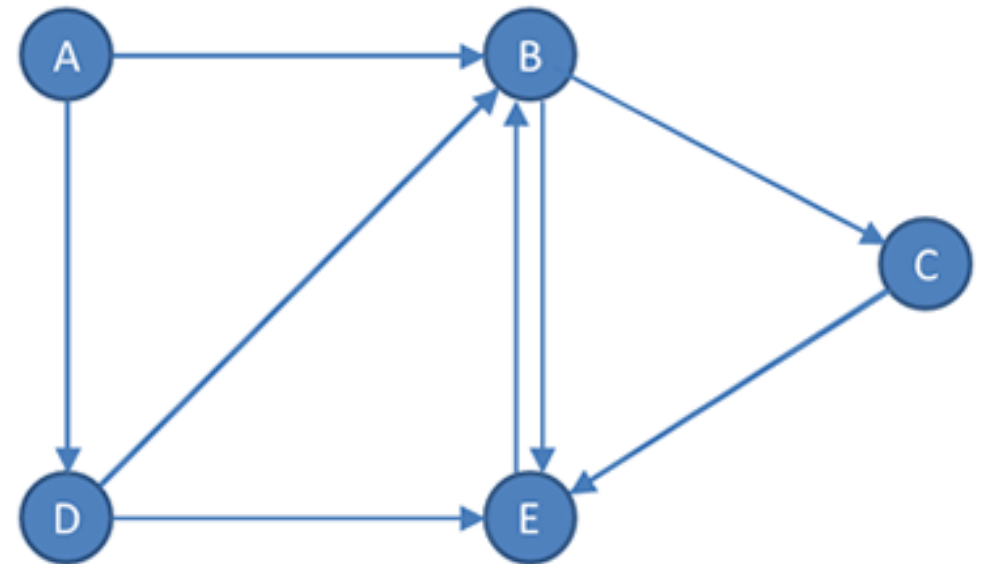


Graph Neural Networks

What is a Graph?

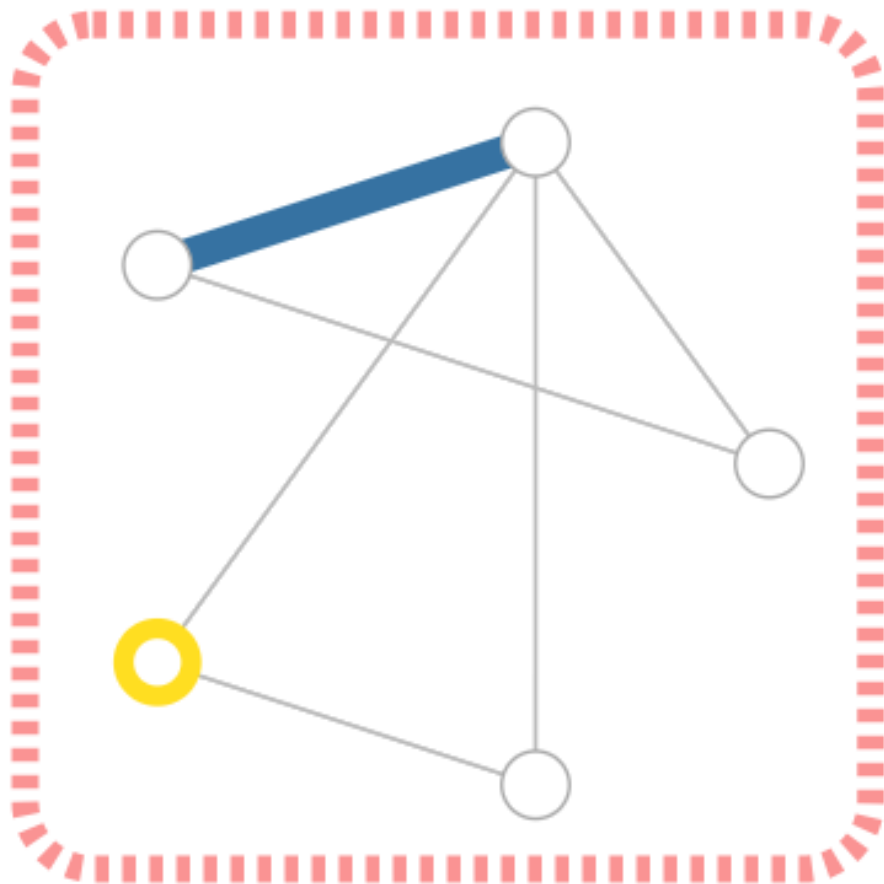


Graph components



Directional Graph example

Graph representation



Vertex (or node) embedding



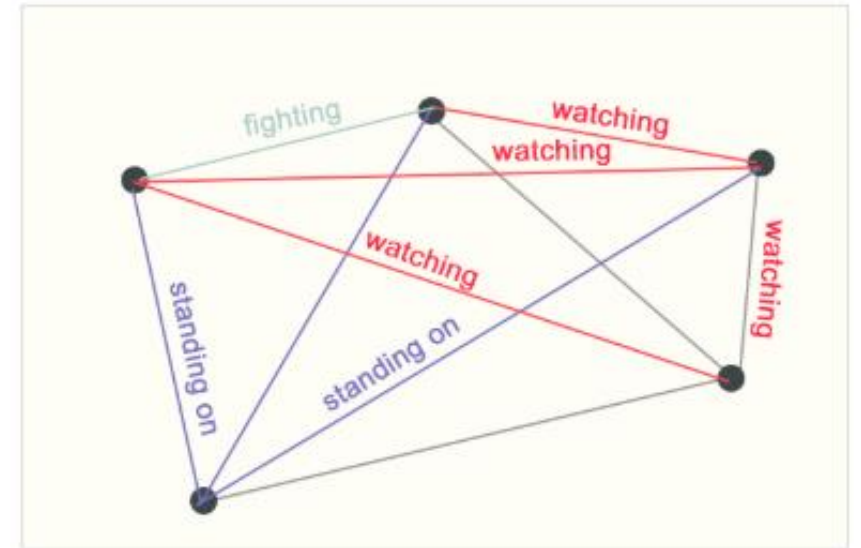
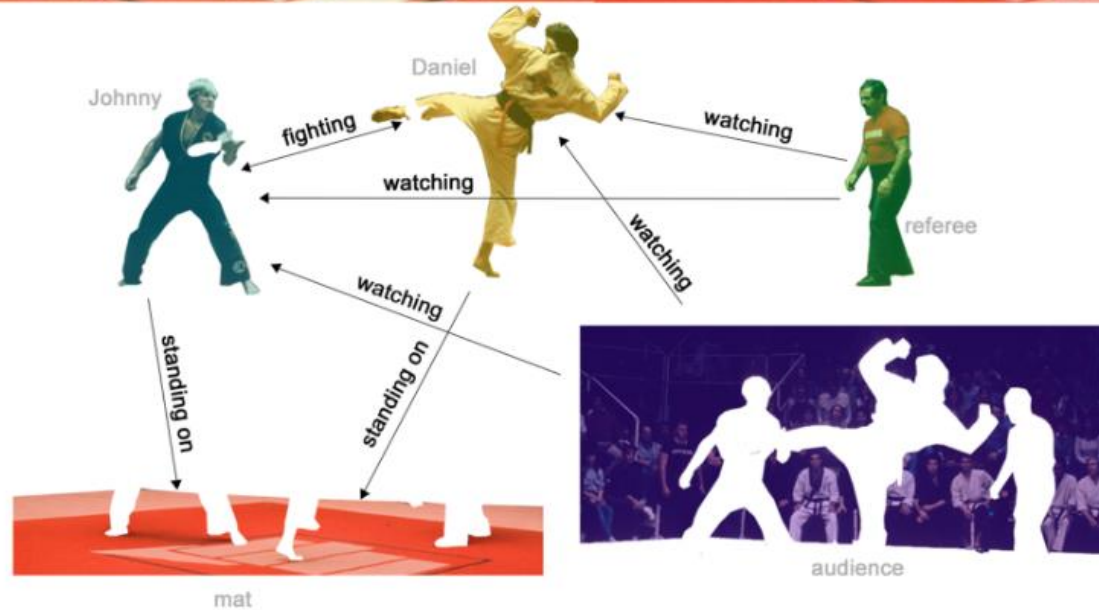
Edge (or link) attributes and embedding



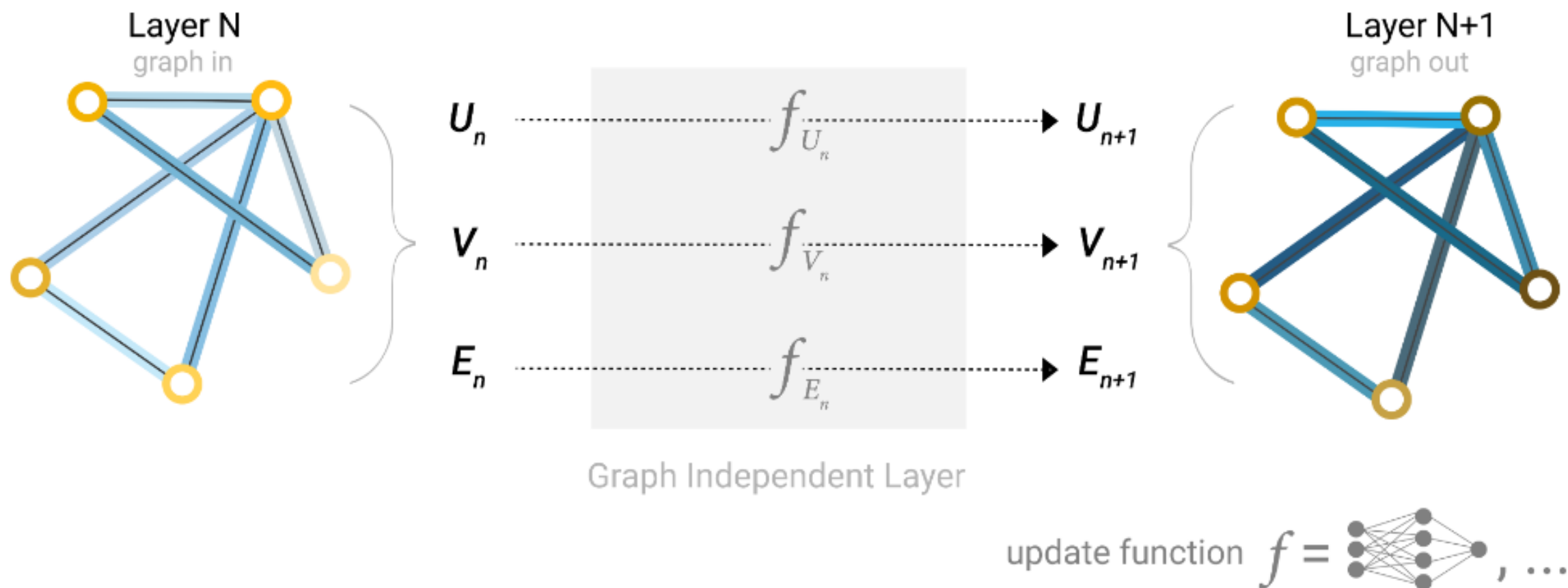
Global (or master node) embedding



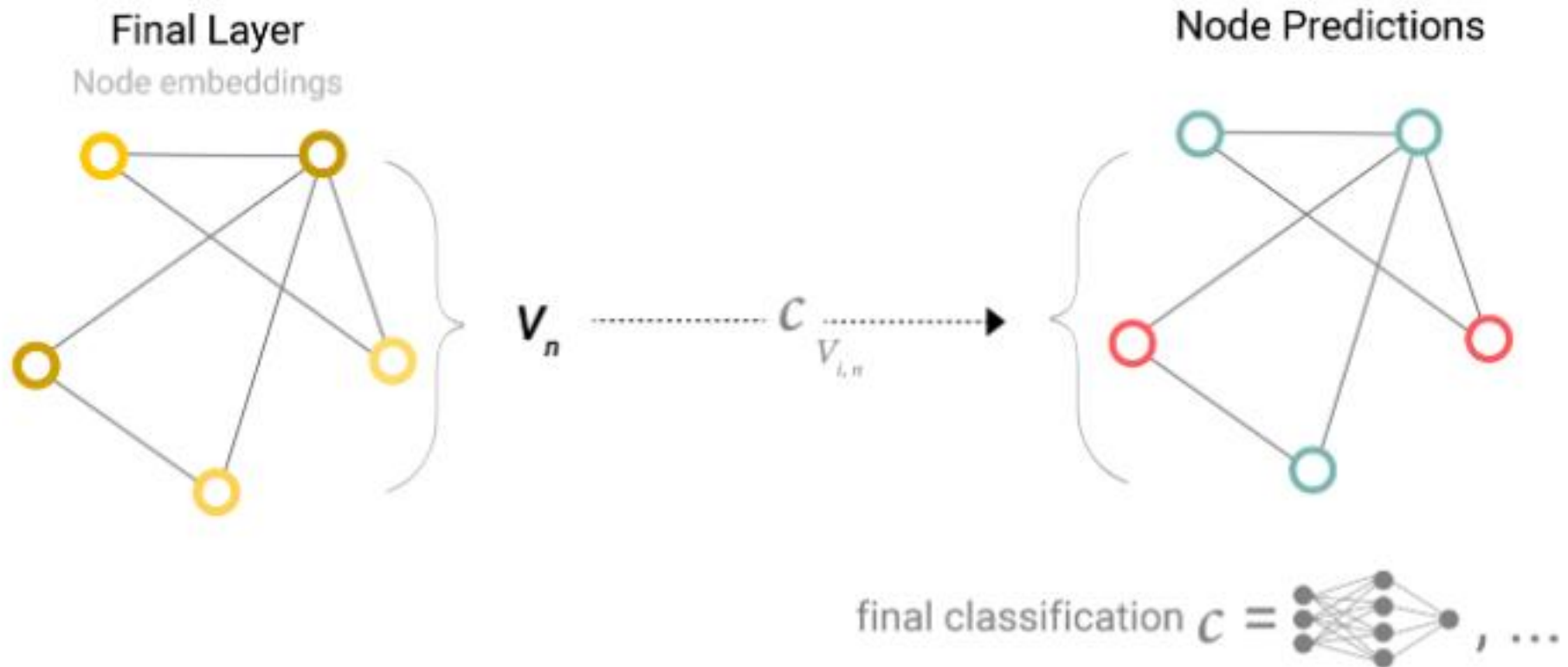
Graph representation



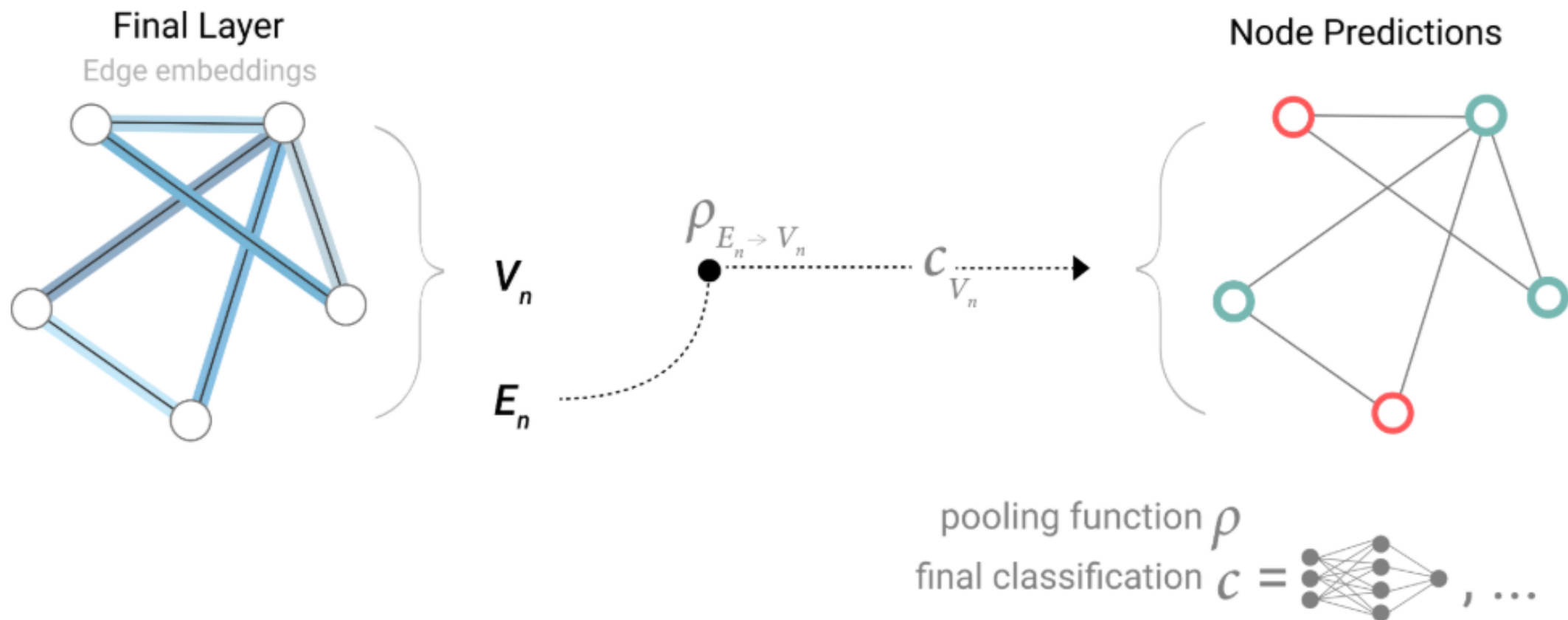
Processing methods: GNN



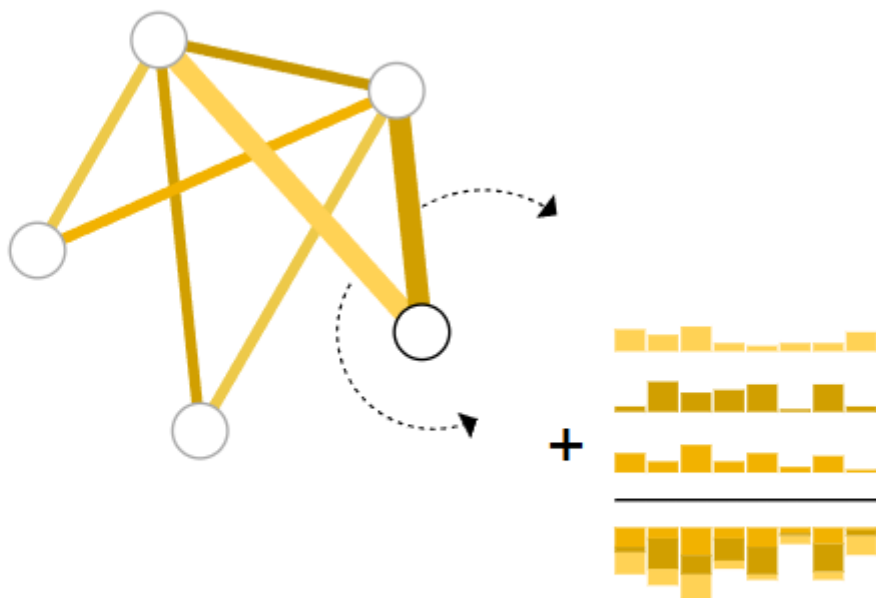
Processing methods: GNN



Processing methods: GNN



Processing methods: GNN

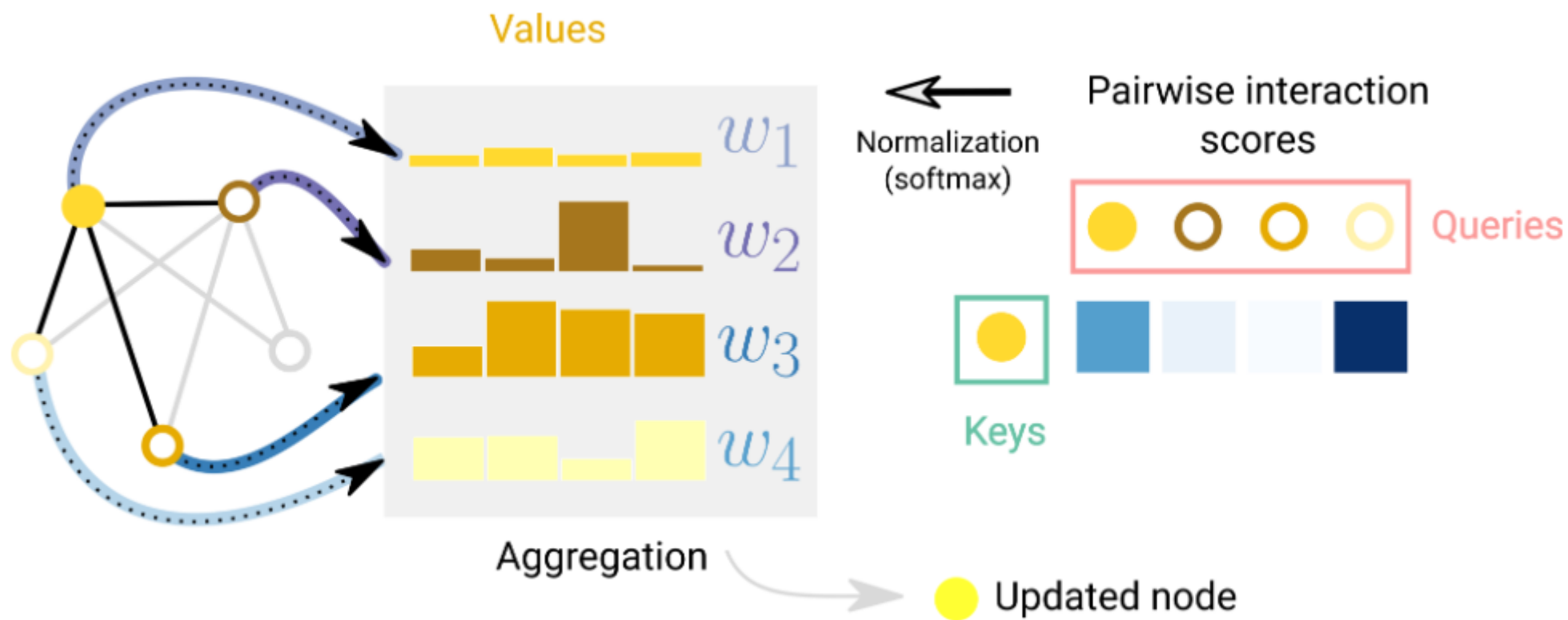


Aggregate information
from adjacent edges

`torch.mean`
`torch.sum`
`torch.max`
`torch.min`

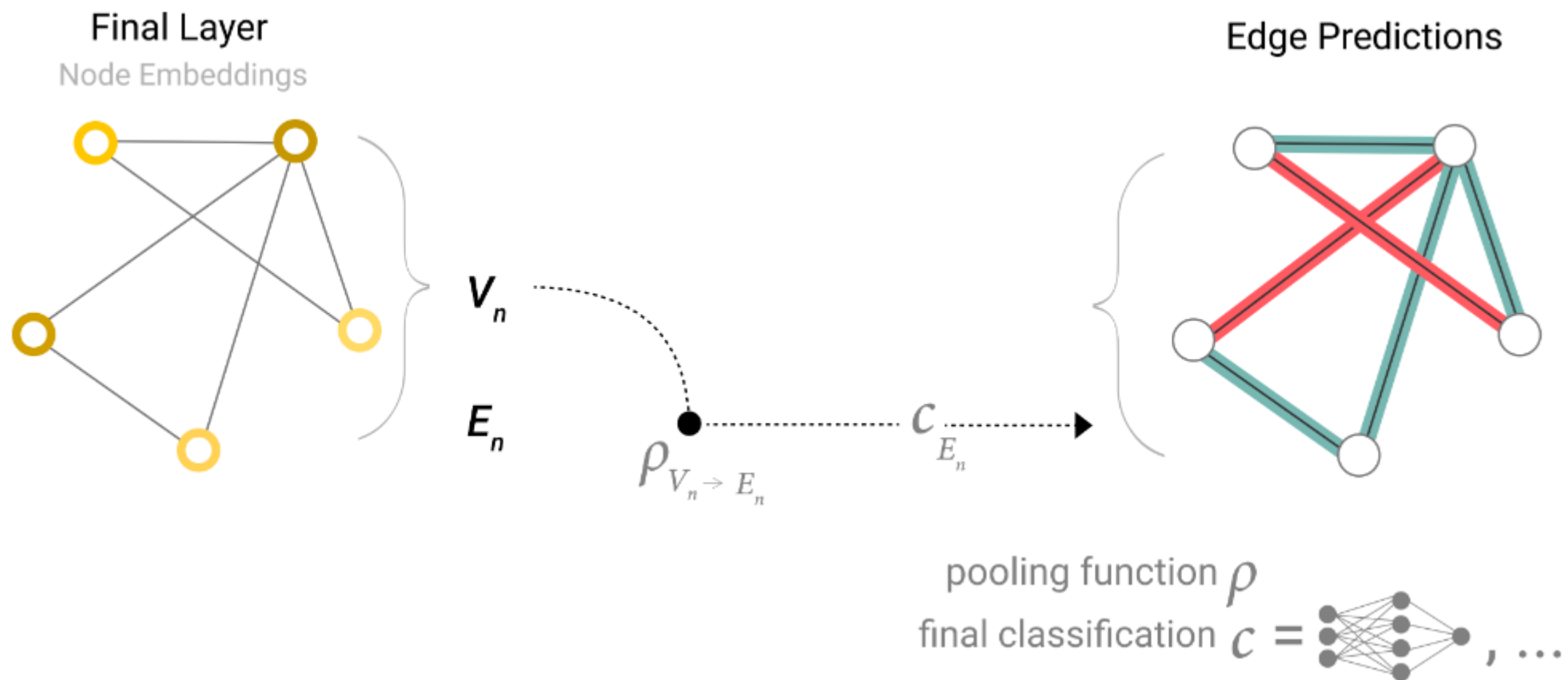
Aggregation methods

Processing methods: GNN

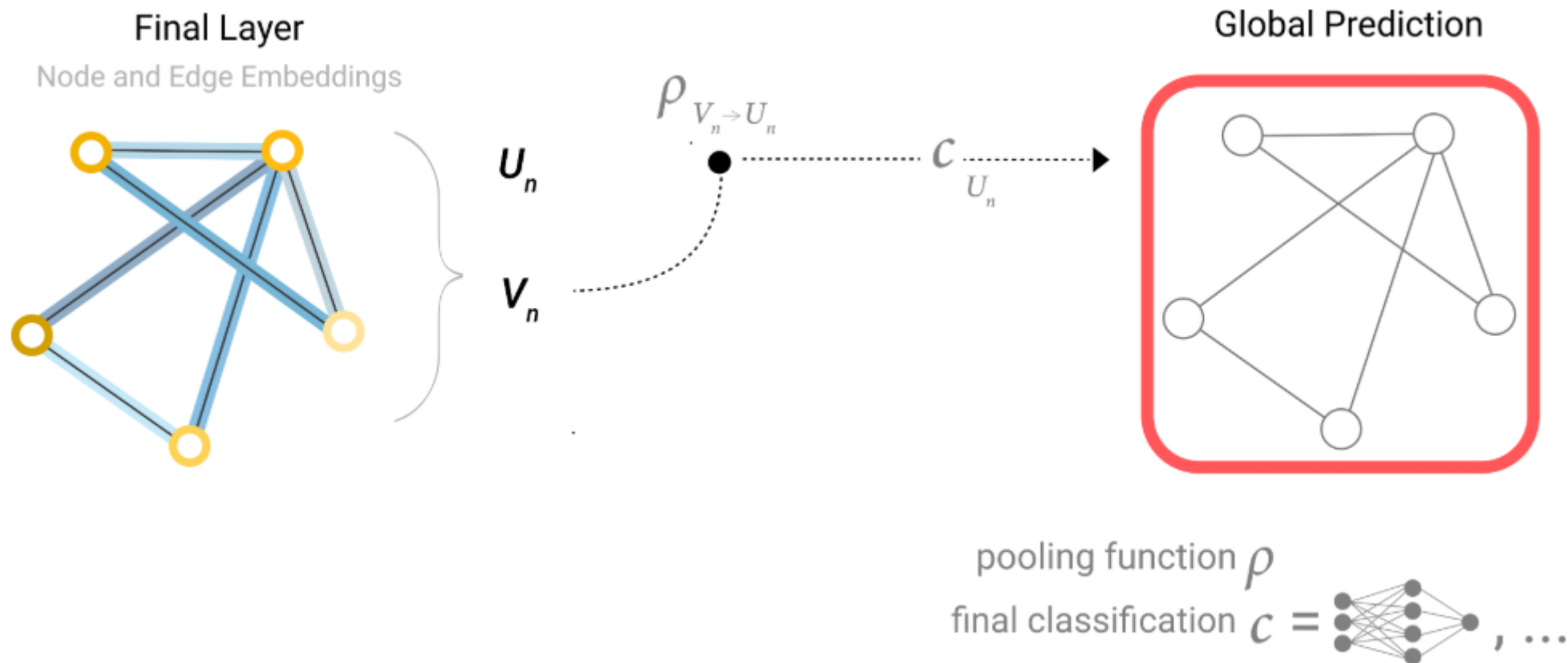


Schematic of attention over one node with respect to its adjacent nodes. For each edge an interaction score is computed, normalized and used to weight node embeddings.

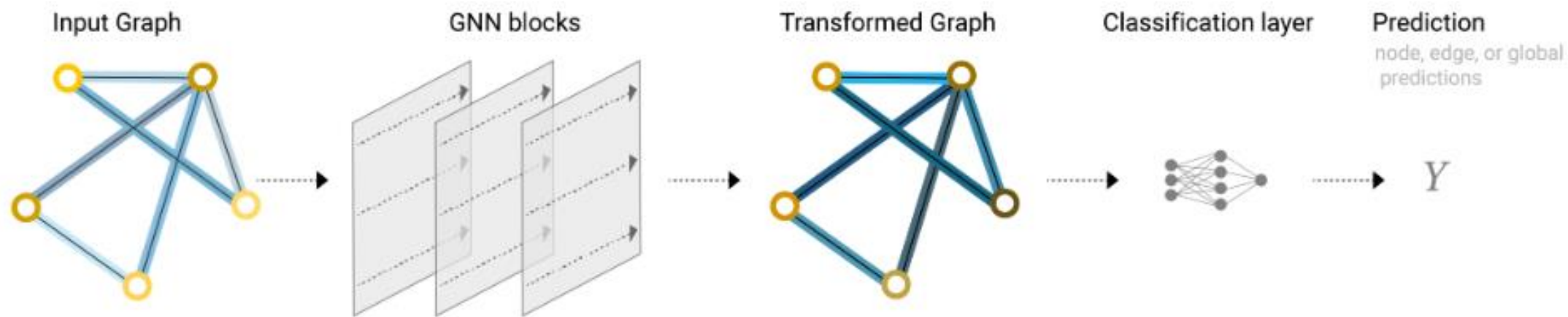
Processing methods: GNN



Processing methods: GNN

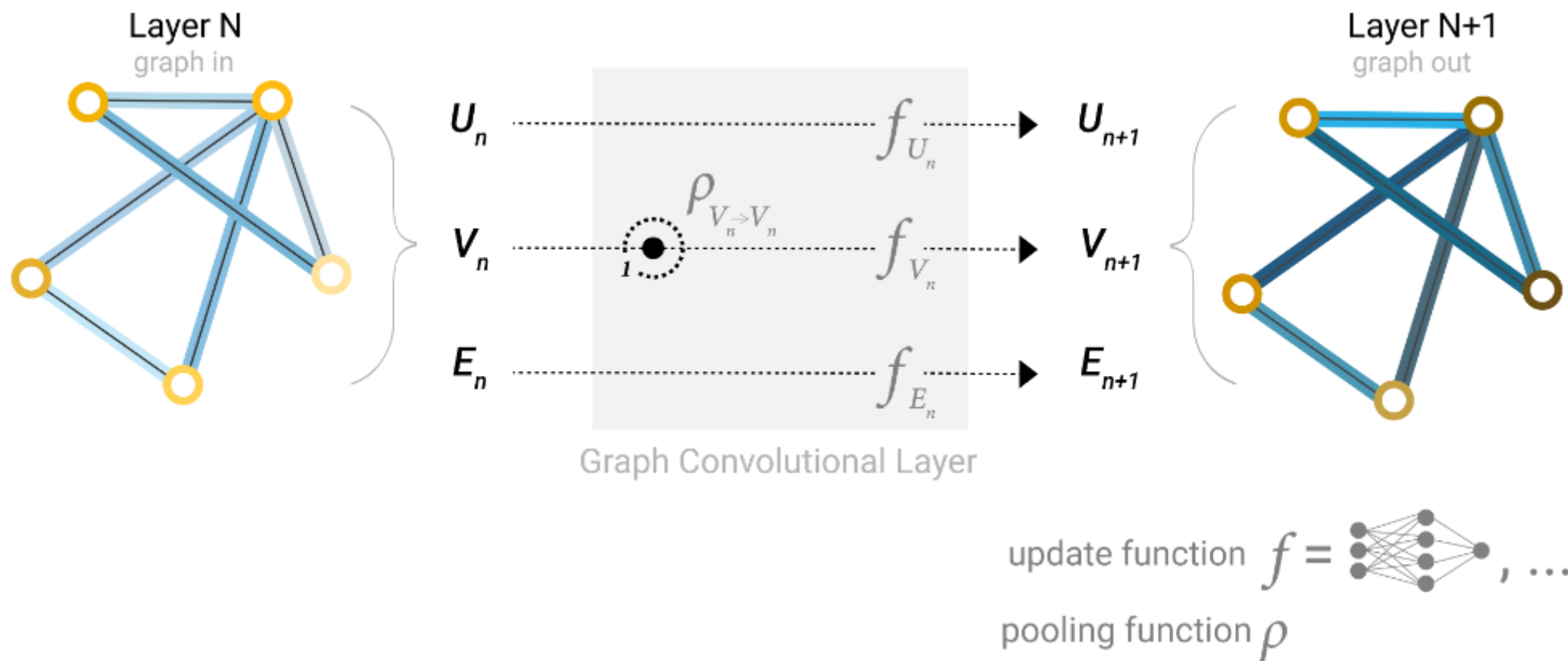


Processing methods: GNN

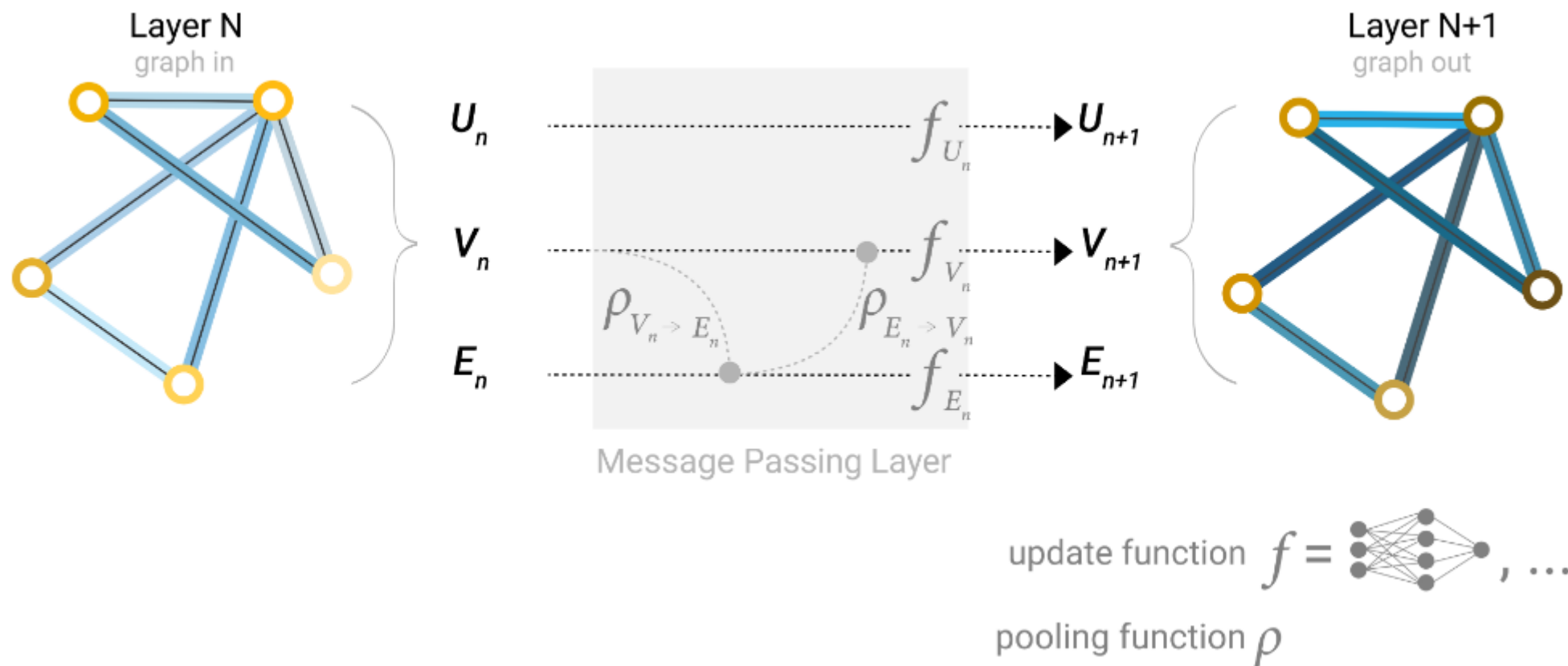


An end-to-end prediction task with a GNN model.

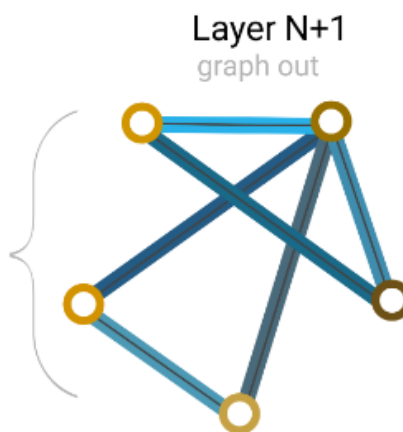
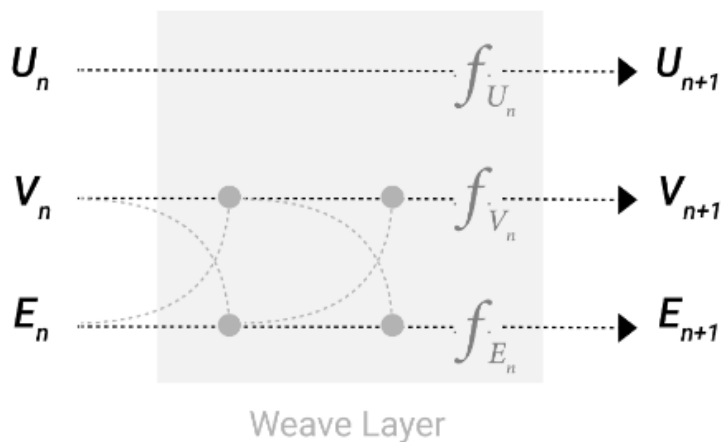
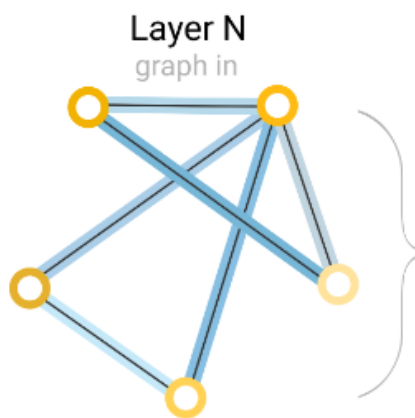
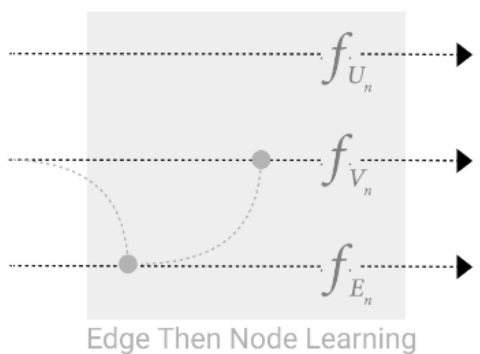
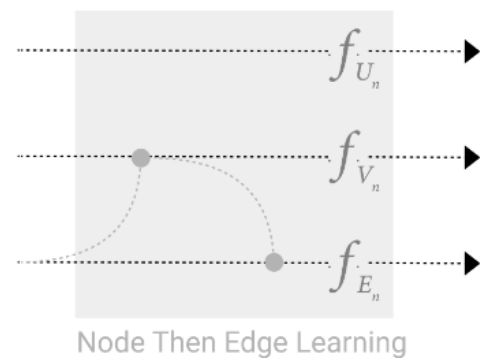
Processing methods: GNN



Processing methods: GNN

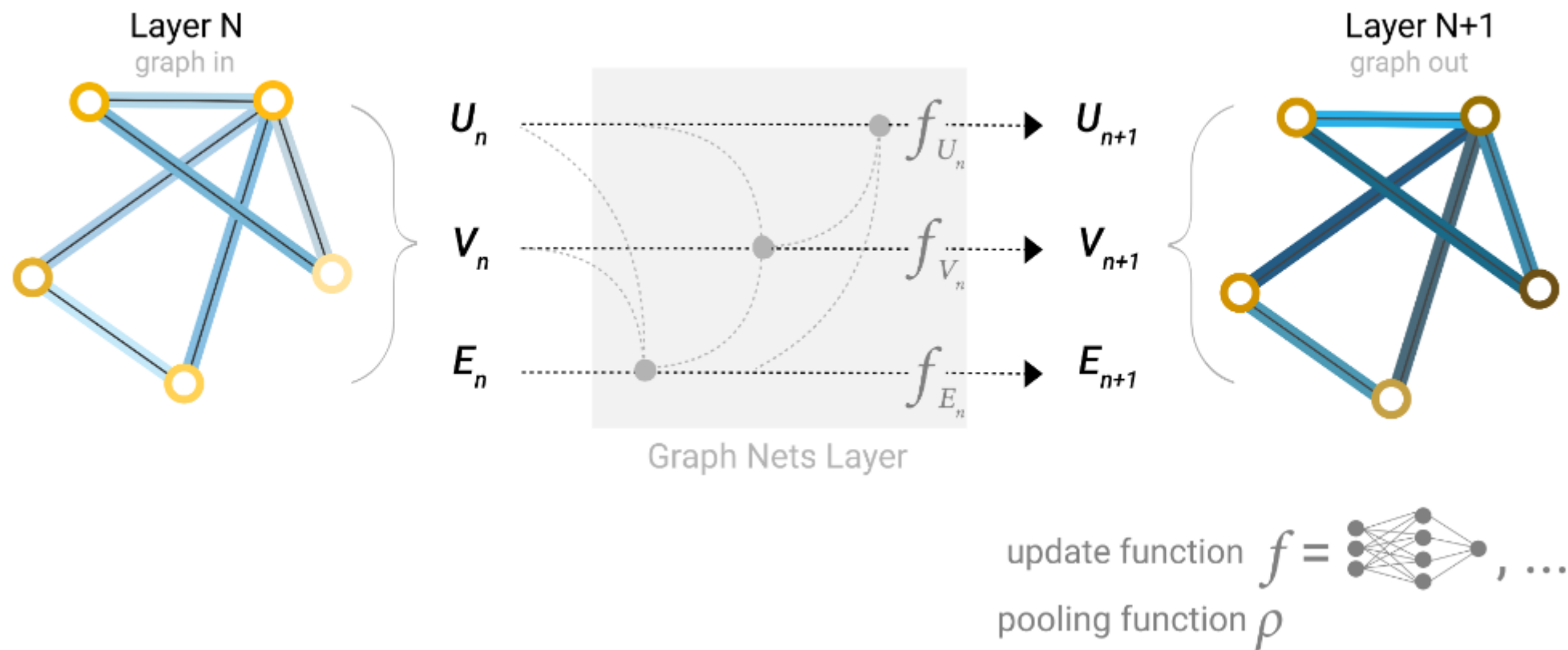


Processing methods: GNN

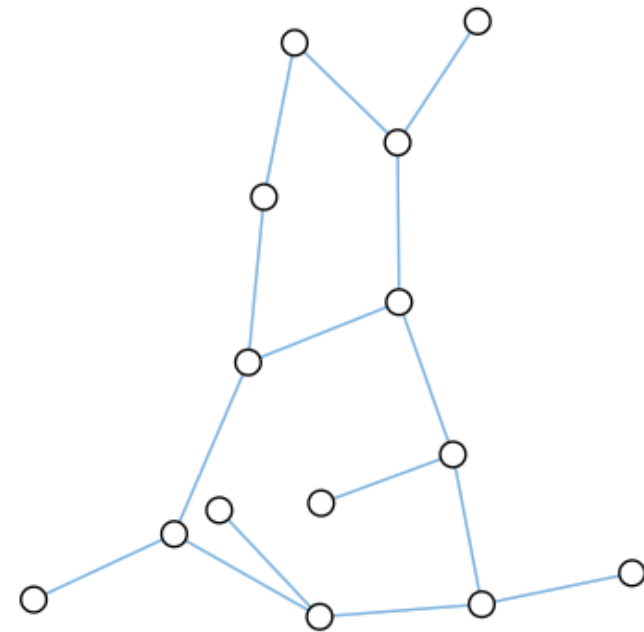
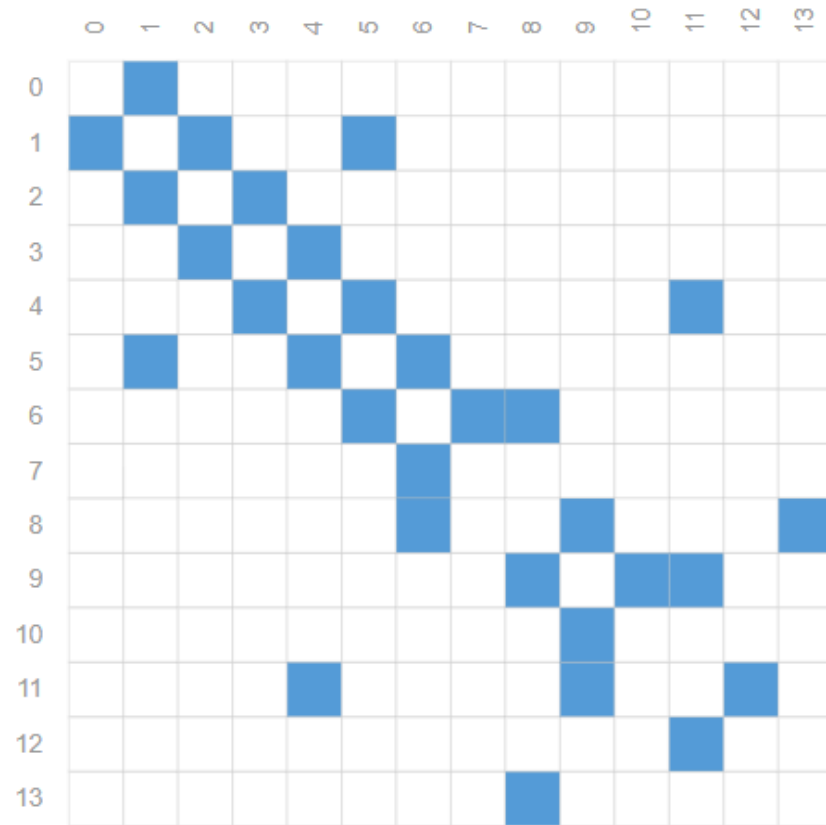


update function $f = \text{[neural network icon]}, \dots$
pooling function ρ

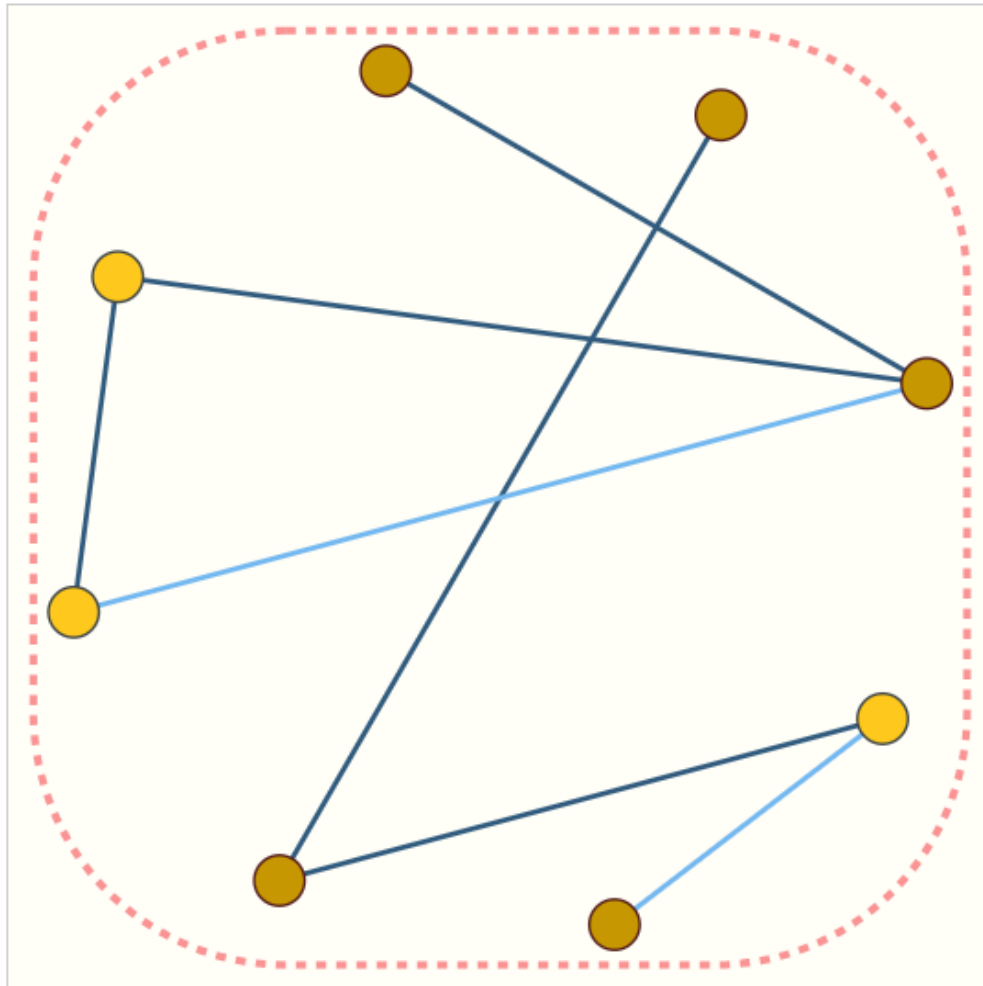
Processing methods: GNN



Talking about adjacency



Talking about adjacency



Nodes

[0, 1, 1, 0, 0, 1, 1, 1]

Edges

[2, 1, 1, 1, 2, 1, 1]

Adjacency List

[[1, 0], [2, 0], [4, 3], [6, 2],
[7, 3], [7, 4], [7, 5]]

Global

0

Adaptive GNN – adjacency layers

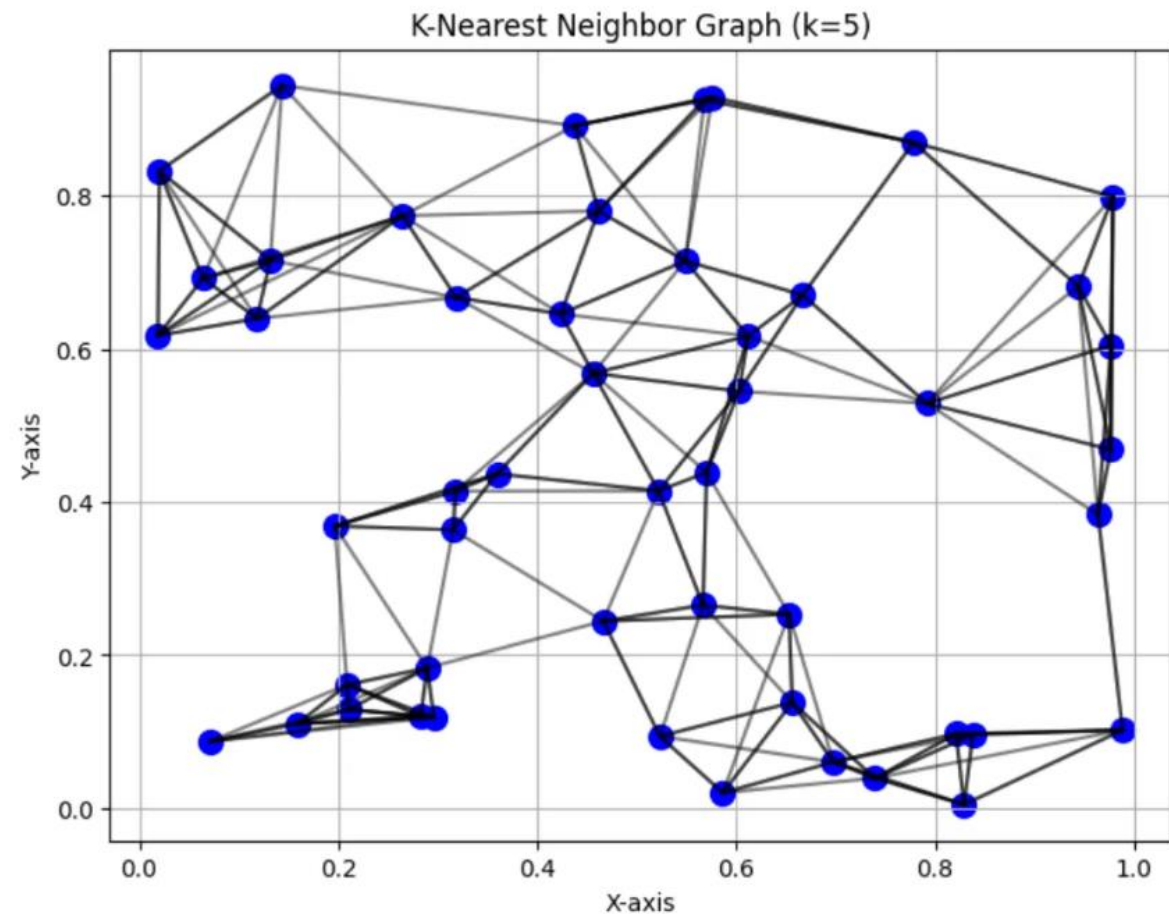
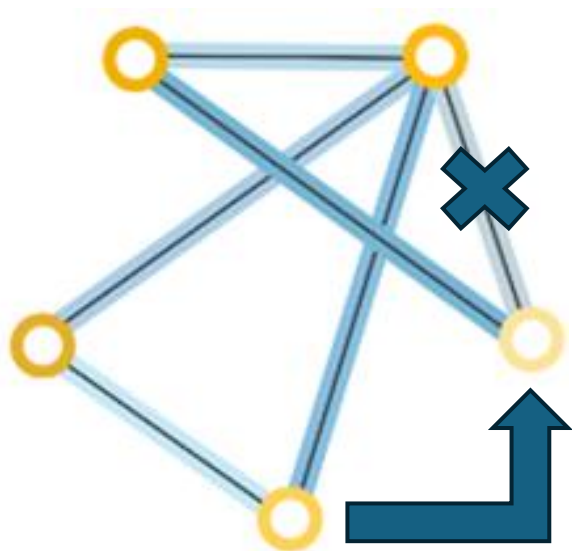
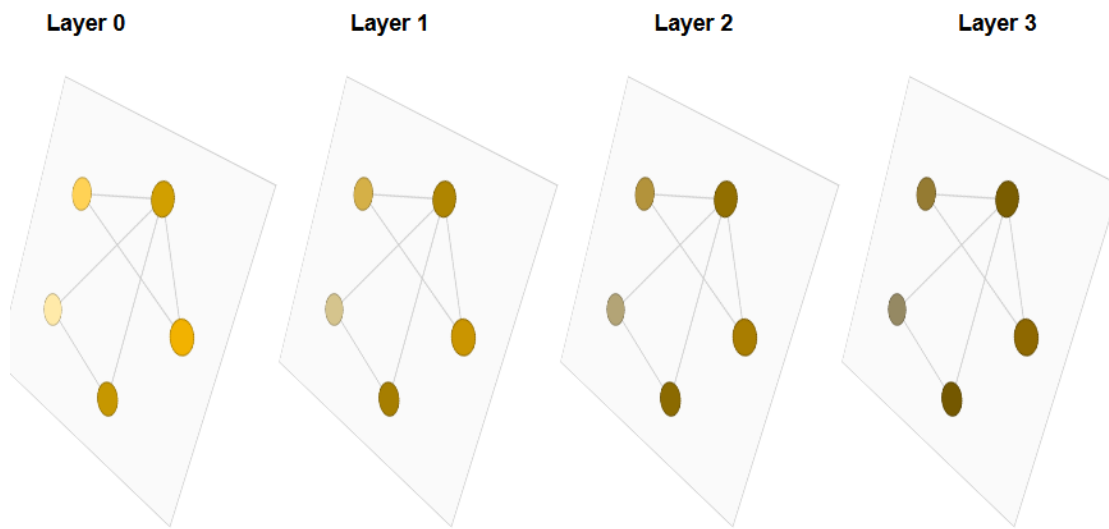


Fig 2: A 5-NN graph obtained from randomly initialized data with 2 attributes. Code generated by ChatGPT 3.5.

Original work

A Gentle Introduction to Graph Neural Networks

Neural networks have been adapted to leverage the structure and properties of graphs. We explore the components needed for building a graph neural network - and motivate the design choices behind them.



<https://distill.pub/2021/gnn-intro/>